

RIDER: Proactive and Reactive Approach for Urban Traffic Management in Vehicular Networks

Thiago S. Gomides^{1,2}, Robson E. De Grande², Fernanda S.H. Souza¹, Daniel L. Guidoni¹

¹Department of Computer Science, Federal University of São João del Rei, Brazil

²Department of Computer Science, Brock University, Canada

Email: {gomides, fsumika, guidoni}@ufs.edu.br, rdegrande@broku.ca

Abstract—Road capacity infrastructure and temporary interruptions in trips constitute the main reasons behind the traffic jam phenomenon. City urbanization and growth further intensify these two reasons through the increase of work area and the demand for mobility. In such a scenario, several issues can emerge, such as higher mobility costs, more frequent traffic jams, more significant environmental damage, reduced quality of life, and more pollution. Therefore, this work presents a Proactive and Reactive Approach for Urban Traffic Management in Vehicular Networks, RIDER, to minimize traffic congestion. RIDER is a fully-distributed protocol that can assume proactive and reactive behaviors for sharing traffic condition information. Vehicles with traffic condition information can organize themselves to improve traffic flow and reduce traffic congestion. In the proposed solution, vehicles monitor the road traffic condition and proactively share this information when needed, considering adaptive multi-hop communication. If vehicles do not have nearby road traffic information, they executed a reactive traffic information discovery. RIDER was evaluated and compared to previous works, regarding the number of transmitted messages, packet collisions, and traffic congestion metrics.

Index Terms—Vehicular ad-hoc Networks, Traffic Congestion, Data Dissemination Protocols

I. INTRODUCTION

Mobility control and planning are critical challenges in the centers of dense urban areas, especially because of the urbanization during the last decades. Thus, the concentration of people in large cities increased due to essential day-lives activities, such as work and education. The World Urbanization Prospects of the United Nation (UN) researchers estimated that more than two-thirds of the world population will live in urban areas by 2050, expanding the challenges related to mobility control [1]. The lack of urban planning rises mobility issues as more people use private transportation, thus contributing to the emergence of congestion. In economic terms, the American Transportation Research Institute estimates that the cost of transit in the USA exceeds \$74.5 billion annually [2].

Technological solutions have become popular to reduce traffic congestion in urban centers, due to lower implementation costs and influencing the lives of people. Urban Traffic Congestion Systems use information and communication solutions that consider the characteristics of roads and users to provide mechanisms to improve the use of the available urban road infrastructure. In general, these systems assess the traffic condition, detect congestion/traffic jams, and suggest alternative routes with better travel time with a small increase in the route

distance. The use of Vehicular Ad-hoc NETWORKS (VANETs) stands out as an essential component in the development of vehicular congestion management systems and the design of future easy-to-use solutions.

The use of VANETs contributes to this process with the incorporation of sensing technologies, wireless communication interfaces, storage, and processing capacity in vehicles [3]. Vehicles with communication and sensing capabilities can monitor and share the conditions of their displacement, considering the problems of urban mobility, such as reducing congestion and accidents, decreasing the emission of polluting gases, lessening the travel time, and increasing the vehicle flow. Several previous works, such as [4], [5], [6], [7], [8], and [9], have studied the use of VANETs in the design of solutions to minimize congestion in urban centers. However, these works rely quite heavily on infrastructure devices or use a high number of distributed messages to decrease vehicular traffic congestion.

This work thus proposes a vehicle traffic congestion solution to reduce traffic jams in urban centers. The proposed approach, named RIDER, uses a Proactive and Reactive data dissemination protocols to share traffic information according to the traffic network of urban areas. The proposed solution is based on local, individual vehicle observations to estimate traffic conditions. Whenever a vehicle identifies an increase in traffic congestion during its route, it shares this information using the proposed proactive and adaptive information dissemination protocol. Based on the detected traffic intensity, the vehicle shares this information with its one-hop neighbors or considering a multiple neighborhood of vehicles. Upon the receiving of traffic information, a vehicle evaluates if there is a need to determine alternative routes with less congestion. If the vehicle verifies missing traffic information on nearby roads, it executes the reactive traffic information discovery protocol. We conducted a performance evaluation that compared previous approaches to our proposed solution, which presented better results in terms of transmitted messages and vehicular traffic congestion metrics.

The rest of this work is organized as follows. Section II presents the literature solutions to reduce traffic congestion in urban centers. Section III gives an overview of the proposed solution and describes its design principles. Section IV details the proactive and reactive data dissemination protocols. Section V shows the performance analysis and discusses results.

Finally, Section VI concludes the paper and provides some future work directions.

II. RELATED WORKS

Regarding the communication infrastructure and the vehicular network context, we may classify solutions to decrease urban traffic congestion into categories such as centralized, infrastructure-based and distributed. Centralized approaches a central server with a global view of traffic information is responsible for sending alternative routes with lower congestion rates to vehicles. Such solutions end up demanding high communication cost [9], [10]. In infrastructure-based solutions, the city infrastructure might adapt where supplemental infrastructures such as Road-Side Units (RSU) are placed along the roads to receive, process, and disseminate traffic information and alternative routes. However, RSUs brings an extra cost for a vehicular traffic solution [6], [7]. Distributed solutions rely purely on vehicle communication. Vehicles disseminate and receive traffic information during their trips, aiming to estimate alternative routes locally [5].

A traffic management algorithm named *Random k-Shortest Path* (RkSP) is presented by [11]. Each vehicle has k -alternative paths calculated to obtain a distributed flow of vehicles, and when receiving the paths, the vehicle randomly chooses one of them. Moreover, the difference between the fastest k -paths and the slowest k -paths cannot be higher than 20% to avoid increasing its travel time. The authors present the RkSP to propose DIVERT: a vehicular traffic congestion system to reduce traffic jams [9]. DIVERT consists of a centralized system, which uses road segment travel times to calculate new routes. Therefore, a central server periodically requests the travel time of the road segments from the vehicles. The server can analyze the global traffic flow and send faster new routes to the vehicles. The central server conducts all the data through 4G communication.

A distributed traffic congestion solution for vehicular networks, named FIRE-NRD, analyzes vehicular displacements and allows vehicles to detect and classify congestion in urban centers [12]. Besides, traffic information is collected while vehicles travel their route, and it supports the decision of new route paths. Each vehicle classifies the congestion considering the difference between the travelled distance and the expected distance in free-flow conditions. Whenever the vehicle detects changes in the predicted flow, vehicles should disseminate such information for its immediate neighborhood, helping the choice of the next road decision. When approaching road intersections, each vehicle chooses the next road segment to proceed, and this process repeats along the entire route.

In [5] the authors present a urban traffic management system based on an opportunistic knowledge dissemination to reduce traffic congestion named *Preventing traffic congestion through a fully-Distributed Rerouting Algorithm* (PANDORA). PANDORA employs a floating content approach with traffic information located in critical areas. Critical areas consist of regions where the system should work to reduce traffic congestion. Each critical area contains several sub-regions,

and vehicles moving inside a specific sub-region receive the travel time of all roads within that region. Applying a 2-second *beacon* interval, vehicles share their locations and speed with all vehicles inside the same sub-region to detect congestion and improve their route guidance. Considering the information received and based on the Level-Of-Service (LOS) of the Highway Capacity Manual (HCM) [13], vehicles can calculate faster alternative routes whenever needed. An important parameter for PANDORA consists of the critical area definition since it implies the number of transmitted messages to execute the vehicular traffic management solution.

III. URBAN TRAFFIC CONGESTION SOLUTION

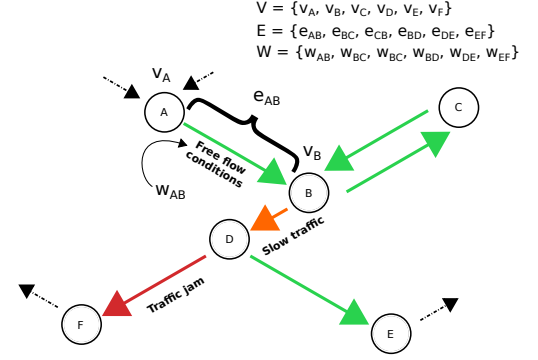
To further reduce the communication overhead that existing traffic congestion solutions generate, we introduce **RIDER**, Proactive and Reactive Approach for Urban Traffic Management using vehicle-to-vehicle (V2V) communication. A **Proactive** and **Reactive** data dissemination protocol enables the sharing of traffic information among vehicles at a minimum communication cost. Proactively, vehicles disseminate information when changes in vehicle flow are detected, and reactively, vehicles can request this information. The proposed vehicular traffic congestion solution addresses the following issues. (i) It enables the local assessment of congestion on the streets/roads where vehicles travel. (ii) It implements a collaborative and adaptive sharing of local perception of vehicular traffic. (iii) It allows estimating alternative routes using only the local information received from other vehicles.

Figure 1(a) illustrates the proposed solution. The vehicles share their local observation of the traffic with the nearby vehicles using V2V communication. Upon receiving a new traffic information message, the blue vehicle can find an alternative route without congestion or with lower congestion, represented by a green line. In our work, vehicles define their local perception of traffic considering their displacement, travelled distance, as originally defined in [12]. In general, the context of the urban traffic congestion environment can be defined as follows:

Definition 1 (Vehicular Environment). *The urban vehicular environment is defined as a weighted and directed graph $G = (V, E)$. The set of $V = \{v_0, v_1, v_2, \dots, v_n\}$ represents the intersection of roads, and the set of $E = \forall (v_i, v_j)$ represents the roads segments among intersections. Each edge $e_{ij} = (v_i, v_j)$ represents the road segment connecting the intersections v_i and v_j with an associated cost w_{ij} , which represents the cost to travel in the segment e_{ij} . The weighting set is represented as $W = \{w_{ij}, i \neq j\}$ and $w_{ij} \rightarrow R_+^*$. Let N be the set of vehicles (nodes) where $N = \{n_0, n_1, \dots, n_{m-1}\}$ and m is the number of vehicles on the environment. Each vehicle implements a database called Road Segment Database, represented by Ψ , to store the observed or received road traffic information. Considering a node n , if $\Psi(e_{ij}) \neq \emptyset$, the node n already received or observed vehicular traffic information about the e_{ij} road segment.*



(a) Urban traffic congestion solution



(b) Graph $G = (V, E)$ representing the road map of Figure 1(a).

Fig. 1: Overview of the proposed solution

In the urban vehicular environment, each vehicle can process, analyze, communicate and making distributed decisions. For this purpose, we assume that all vehicles are equipped with an on-board unit (OBU), IEEE 802.11p communication interface, sensors, global position system (GPS) receiver, the map with the roads, and characteristics of the city. The city road map is represented by the directed and weighted graph $G = (V, E)$, illustrated in Figure 1(b). Besides the traffic conditions, represented by different values of the weight w and displayed with different colours in the figure, the graph G also has attributes, such as street dimensions, number of lanes, traffic light positions, and maximum allowed speed.

While travelling on each road belonging to the route from its origin to destination, vehicles classify changes in traffic conditions by making the association between the **Traveled Distance** of the vehicle and the **Expected Distance** that the vehicle should travel if the road has a free-flow condition. The difference between the two distance estimations defines a factor of congestion, where vehicles that have the travelled distance similar to the expected distance in free flow have a congestion factor of 1. If this difference is higher than 90%, the congestion factor contains value 10. A congestion factor of 10 indicates a fully congested road since the vehicle has made a short displacement on it. Therefore, the congestion factor is an integer from 1 to 10. Whenever the vehicle identifies changes its congestion factor, it executes the **Proactive data dissemination** protocol to send a message containing the estimated congestion factor to neighbouring vehicles.

Each vehicle has a local database to store the received information, and it checks if there exists an alternative route with less congestion when it approaches intersections. If its local database does not contain traffic information for any adjacent road to the intersection under analysis, it executes the **Reactive Data Dissemination** protocol to discover the missing conges-

tion factors. For the calculation of a new route, each vehicle must update $w_{ij} \forall e_{ij} \in G(V, E)$ using the information from its database and considering $w_{ij} = CongestionFactor_{ij} \times ETT_{ij}$, where ETT_{ij} represents the Expected Travel Time of the road e_{ij} under free-flow conditions. Thus, a vehicle calculates the weight w_{ij} by adding a traffic increase factor to the travel time in free-flow condition. Consequently, the vehicle identifies alternative routes by applying a minimal cost path algorithm in $G(V, E)$.

IV. DATA DISSEMINATION PROTOCOLS

A. Proactive Data Dissemination

The Proactive Data Dissemination protocol is responsible for providing and sharing sufficient information for the discovery of alternative routes/paths with lower traffic congestion. Fundamentally, the design of the protocol aims to fit a distributed and highly mobile environment using V2V communication. Algorithm 1 illustrates the proactive protocol. Based on the estimated congestion factor (variable $cf_{e_{ij}}$), the Proactive Dissemination is executed while the vehicle travels (Algorithm 1, line 1) and waits until one of the two behaviors is triggered (i) when there is a change in the flow conditions and (ii) when there is a need to maintain the road traffic information updated. In the first case (Algorithm 1, line 2), each vehicle that detects an increase in flow must proactively disseminate the information according to the Adaptive Information Sharing procedure represented by function $send_{AIS}$ (Algorithm 1, line 4). The procedure $send_{AIS}$ is composed by three parameters where $cf_{e_{ij}}$ is the contention factor, N is the set of vehicles and the function nf returns the number of hops to be reached considering $cf_{e_{ij}}$ and the set of nodes. In addition, when a vehicle performs a $send_{AIS}$ function, the counter t is reset. The Ψ – Road Segment Database contains the road traffic information. Then, the vehicle verifies the need

to keep the road traffic information updated (Algorithm 1, lines 6–13); this verification is important to maintain the road information when the traffic flow does not change under a period T . If the traffic flow does not change and it is not necessary to keep the information updated, the proactive dissemination stops and awaits to repeat the process (Algorithm 1, line 14). The function *wait* returns when there is a change in traffic flow ($cf'_{e_{ij}}$) or when timeout ($t = 0$).

Algorithm 1: Proactive

Input: $\Psi, e_{ij}, cf_{e_{ij}}, T$

```

1 while true do
2   if  $\Psi(cf_{e_{ij}}) < cf_{e_{ij}}$  or  $\Psi(cf_{e_{ij}}) = \emptyset$  then
3      $t \leftarrow T$ ;
4      $send_{AIS}(cf_{e_{ij}}, N, nf(N, cf_{e_{ij}}))$ 
5   end
6   else
7     if  $t = 0$  then
8       if  $e_{ij} \in \Psi$  then
9          $t \leftarrow T$ ;
10         $send(cf_{e_{ij}}, N_{ne=1})$ ;
11      end
12    end
13  end
14   $wait(cf'_{e_{ij}}, t = 0)$ 
15 end

```

The adaptive information sharing procedure executes when the vehicle detects a traffic change or when the vehicle does not have traffic information regarding road e_{ij} in its Road Segment Database (Ψ). The latter case means that it is the first time the vehicle estimates the road traffic information. The procedure has an adaptive coverage area – its coverage depends on the intensity of traffic. If the vehicle estimates a congestion factor of [1 or 2], it disseminates the information using a $TTL = 1$ – a time to live of one hop. If the congestion factor grows, the traffic information must reach greater distances to enable vehicles finding alternative routes since the congestion may be occurring in a region and not just on the road. Consequently, when the congestion factor is [3 or 4] $\rightarrow TTL = 2$, [5, 6 or 7] $\rightarrow TTL = 3$ and [8, 9 or 10] $\rightarrow TTL = 4$. Through extensive experimental analysis, we verify that the vehicles are not able to find better alternative routes considering a neighborhood greater than 4 hops in the information sharing procedure. Another factor considered in adaptive dissemination consists of the number of messages: a lower congestion factor requires less intense dissemination, requiring fewer messages and being limiting the area of dissemination.

Figure 2(a) illustrates different coverage areas of the adaptive information dissemination procedure. Considering the red vehicle positioned in the center of the figure, $TTL = 1$ represents the first neighborhood/hop, $TTL = 2$ designates the second neighborhood. For values of $TTL \geq 2$, it is necessary to create a message re-transmission mechanism to reduce the

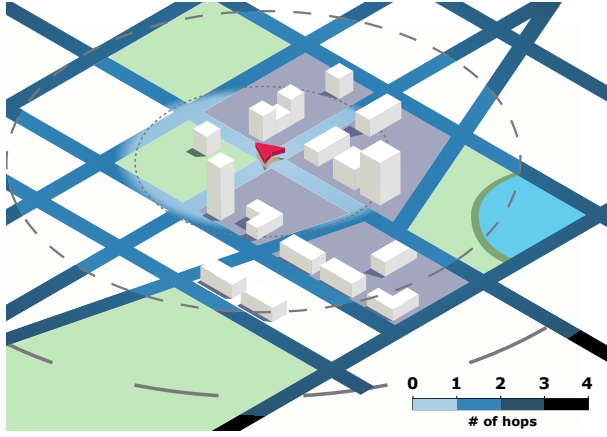
issue of *Broadcast storm*. We define the destination region as angular coordinates centred on the vehicle that is sharing its traffic information. This process is illustrated in Figure 2(b). Only vehicles within the angular intervals can forward the message where the re-transmission area is divided, considering the city map and the number of adjacent roads. The angular regions change based on the layout of the roads because vehicles travel only within urban roads. For each adjacent road, we consider an angular region of 45° to limit data re-transmission. To better control the transmission problem, only one vehicle within each angular region forwards the message. For this, the vehicle that has a greater distance from the sender forwards the message first, cancelling the other transmissions. It is important to note that a message from a vehicle within an angular region does not cancel the forwarding procedure of another vehicle positioned in another region. In this way, the protocol minimizes the overhead of accessing the wireless medium as only one vehicle from each angular region forwards the message, reducing the number of duplicate messages.

The goal of adaptive sharing is to minimize the number of messages to create a distributed information about the traffic conditions. However, a trade-off between information sharing and information lifetime is crucial. Even when traffic congestion does not change, it is necessary to share the traffic information in order to update vehicles that do not have road traffic information yet. One example of this case is a vehicle that does not receive an update when the traffic flow changes, but it is approaching the congested road and should be updated. To prevent traffic information from becoming extinct, we use a timer mechanism. When the vehicle starts moving on a new road, an information-maintenance timer starts every T units of time. If the vehicle detects a congestion factor growth, it executes adaptive information sharing procedure; thus, the vehicle re-starts its timer (Algorithm 1, line 4) since the traffic information is still ongoing. When the timer ends, the vehicle checks whether it has the road information in its database; it then disseminates the information to its neighbors and resets its timer to keep the road information up to date (Algorithm 1, lines 6–9). It is important to emphasize that information maintenance occurs only when there is no change in the congestion factor shared by the vehicle and after T units of time. The information is also disseminated only in the first neighborhood to reduce access to the wireless medium and the number of transmitted messages.

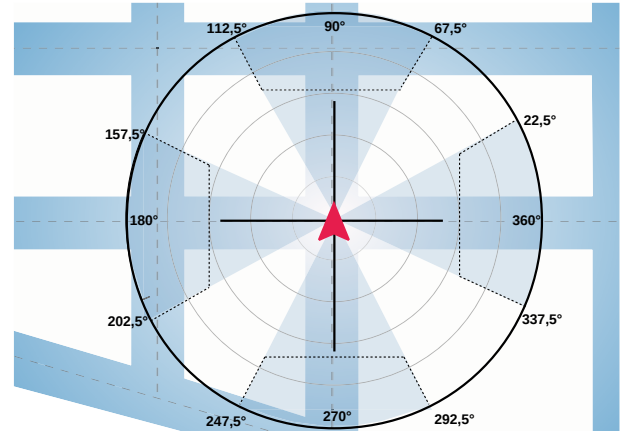
B. Reactive Data Dissemination

The traffic congestion approach executes the reactive dissemination protocol when failures occur in the proactive data dissemination process. Several factors can cause these failures, such as packet collisions due to network overload (different applications using the wireless communication medium), sparse or disconnected networks, and signal attenuation. Whenever necessary, vehicles can request congestion information for vehicles on adjacent roads.

The reactive request information protocol is divided into the request, receive, and response steps. Algorithm 2 shows



(a) Maximum expected range for the multi-hop information dissemination



(b) Analysis in the first hop

Fig. 2: Adaptive Information Sharing

the three functions to implement the protocol. When the vehicle approaches a road junction, it executes the function *request*, illustrated in Algorithm 2, lines 1–5. For the next road segment e_{jl} in its database without traffic information, the vehicle creates and sends a request message to vehicles within the specific road. This message is a congestion level request directed at nodes traveling on the e_{jl} road segment (Algorithm 2, lines 3). Upon receiving a message, vehicles execute the *receive* function (Algorithm 2, lines 6–13). If the received message is a traffic information request from the same node, the vehicle verifies if it is inside the destination road. This verification is essential since the request is performed using the wireless broadcast channel and vehicles in different roads will receive the request message.

Algorithm 2: Reactive

Input: Ψ, e_{ij}

```

1 Function request() is
2   forall  $\bigcup^{vi} e_{jl} \in \Psi$  and  $\Psi(cf_{e_{ij}}) = \emptyset$  do
3     | send(  $N \in e_{jl}, req(cf_{e_{jl}})$  )
4   end
5 end
6 Function receive(msg) is
7   if  $msg = req(cf_{e_{jm}})$  then
8     | if  $e_{jm} \in \bigcup^{vi} e_{jl}$  then
9       |  $msg_{dl} \leftarrow sched_t(\frac{1}{dt_{e_{jm}}}, cf_{e_{jm}}, n_{e_{ij}})$ 
10    | end
11  else if  $msg = resp(cf_{e_{jm}})$  then
12    |  $msg_{dl} \leftarrow \emptyset$ 
13  end
14 Function response() is
15  | send(  $msg_{dl}, cf_{e_{jm}}, n_{e_{ij}}$  )
16 end

```

Upon receiving the traffic information request, each vehicle inside the destination road schedules a response message. The

response scheduling with a delay that follows an ordering that is inversely proportional to the vehicle time in the road. Vehicles with the longest time thus execute the function *response* first (Algorithm 2, line 14). The first vehicle to reply to the request is the one with the best knowledge about the traffic conditions of the road. When receiving the response message, the other vehicles cancel their schedules (Algorithm 2, lines 16–18). The *response* function, illustrated in Algorithm 2 (lines 20–26), creates a new message containing the destination, which is the requesting vehicle, the message type, and the current congestion factor of the road. It is important to point out that when a vehicle close to the requesting one receives the response, it stores the road traffic information in its database, cancelling a possible subsequent request.

V. PERFORMANCE ANALYSIS

The proposed solution is evaluated through communication and mobility simulation in a vehicular environment based on real-world features and parameters. The following tools were adopted: SUMO 0.25 (*Simulation of Urban MObility*) [14] to create and manage the vehicular mobility, OMNET++ 5.1.1 [15] and Veins 4.6 [16] as a vehicular communication *framework* applying the 802.11p standard. The simulations used, as context urban scenario, a region of Los Angeles-USA metropolitan area, obtained through OpenStreetMap¹ [17] tool, with 25 km² as showed in Figure 3.

For comparison purposes, our performance analysis compared the proposed solution with PANDORA, FIRE-NRD, DIVERT and OVM (Original Vehicle Mobility Trace). In the latter, vehicle demands (origin/destination) follow a random definition: vehicle routes take the shortest path and do not change during the simulation. The evaluation used different

¹OpenStreetMap provides road dimensions, directions, traffic lights, speed limits, vehicular preferences, among others.

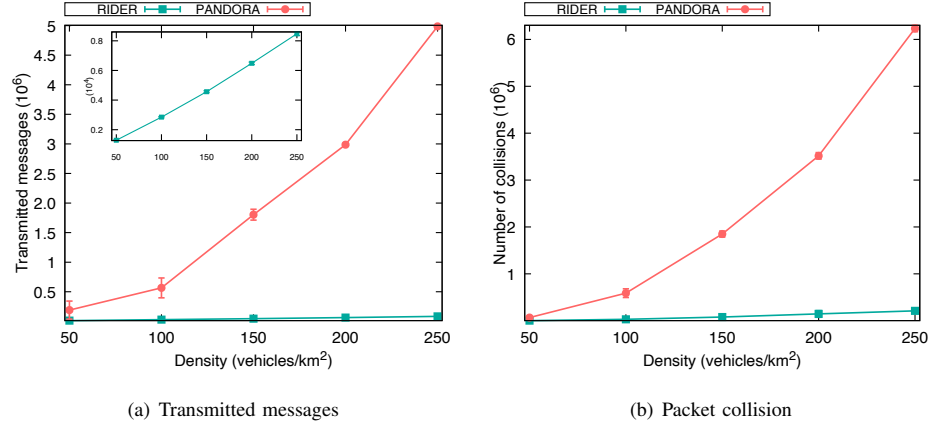


Fig. 4: Network Metrics

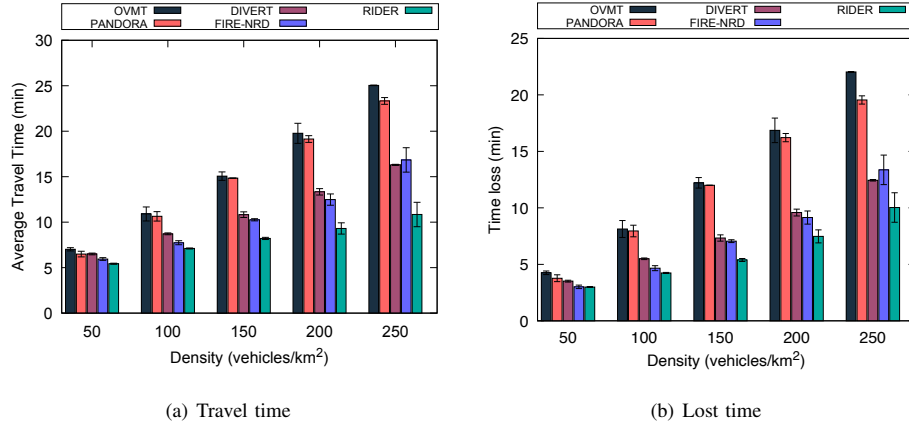


Fig. 5: Vehicular traffic congestion metrics

solutions (PANDORA–OVMT, DIVERT–OVMT, FIRE.NRD–OVMT and RIDER–OVMT) and the y-axis represents the cumulative percentage of vehicles that satisfy the ratio. Figure 6(a) shows the relative travel time considering a density of vehicles equals to 100 vehicles/km² with a lower number of congested roads. DIVERT, which has a global view of traffic congestion, better organizes the flow of vehicles to decrease travel time. It is possible to verify that 70% of the vehicles that execute DIVERT have a travel time lower than $0.5 \times \text{OVMT}$ (relative travel time ≤ 0.5). PANDORA, DIVERT and the proposed solution have 80% of vehicles with a travel time lower than OVMT (relative travel time ≤ 1), where FIRE-NRD has only 70% of vehicles. It is important to note that the solutions are not capable of reducing the travel time of all vehicles. The estimation of traffic congestion of all solutions may be insufficient to verify all congestion points in the scenario. In this case, alternative routes may be more congested than the previous one, increasing the travel time. However, on average, the travel time is decreased, as we can observe in Figure 5.

Figure 6(b) shows the lost time for all solutions. DIVERT

also presents better results considering relative lost time compared to the other solutions, and 80% of the vehicles have a travel time lower than $0.5 \times \text{OVMT}$, where RIDER, FIRE-NRD and PANDORA achieve 38%, 15% and 3%. DIVERT, RIDER, PANDORA and FIRE-NRD have, respectively, 86%, 77%, 75% and 50% of the number of vehicles with a lost time compared to OVMT.

When vehicle density is 250 vehicles/km², as illustrated in Figure 7, the proposed solution presents better results compared to the other solutions. In the case of a large number of congestion spots in the urban area, the global view of traffic information is not sufficient for DIVERT overtaking the results of the proposed solution. Thus, the traffic classification, together with the proposed Proactive and Reactive data dissemination protocols, can reduce travel time and lost time in congestion-intensive scenarios.

VI. CONCLUSION

In this paper, we proposed an adaptive and distributed traffic congestion solution system for vehicular networks to reduce traffic jams in urban centers. The proposed solution employs

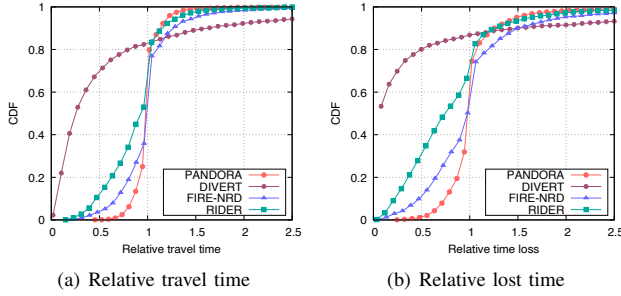


Fig. 6: Network density of 100 vehicles/km²

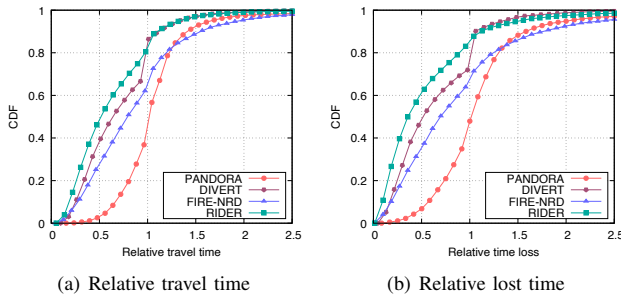


Fig. 7: Network density of 250 vehicles/km²

the vehicle to vehicle communication and can create a local database containing traffic information. The main goal of the proposed solution, named RIDER, is to reduce the vehicle average travel time with a low impact in the number of transmitted messages in urban traffic congestion. RIDER solution relies on proactive and reactive data communication protocols. During its route, each vehicle monitors the traffic congestion of its current road. When changes are detected, the vehicle proactively and adaptively shares its estimated congestion information with nearby vehicles. If the estimated congestion is intense, more vehicles should receive this information in order to find alternative routes, since the traffic jam may impact the routes of several vehicles. In case the vehicle does not have traffic information for nearby roads, it performs reactive information discovery. Using all the received information, the vehicle checks for alternative routes with less congestion. The simulation results show that the proposed solution is capable of reducing congestion, reducing the average travel time and the lost time by vehicles. The proposed solution also reduces the number of messages to reduce traffic congestion in urban environments. As future work, we plan to investigate statistical algorithms for the discovery of alternative routes in order to provide a better load balance with new paths.

ACKNOWLEDGMENT

The authors would like to thank The Minas Gerais Research Foundation (FAPEMIG) under Grant No.: APQ-03120-17, Canadian Bureau for International Education (CBIE) through The Emerging Leaders in the Americas Program (ELAP),

The Brazilian National Council for Scientific and Technological Development (CNPq) under Grant No.: 150545/2018-5. This study was financed in part by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - Brasil (CAPES) - Finance Code 001. This work is partially supported by NSERC Discovery grant.

REFERENCES

- [1] U. DESA, "World urbanization prospects: The 2018 revision (st/esa/ser. a/420)," 2019.
- [2] G. Cookson and B. Pishue, "Inrix global traffic scorecard," *INRIX research*, 2019.
- [3] E. Ahmed and H. Gharavi, "Cooperative vehicular networking: A survey," *IEEE Transactions on Intelligent Transportation Systems*, vol. 19, no. 3, pp. 996–1014, 2018.
- [4] S. Wang, S. Djahel, and J. McManis, "A multi-agent based vehicles re-routing system for unexpected traffic congestion avoidance," in *17th International IEEE Conference on Intelligent Transportation Systems (ITSC)*, Oct 2014, pp. 2541–2548.
- [5] A. M. de Souza, N. L. S. da Fonseca, and L. A. Villas, "A fully-distributed advanced traffic management system based on opportunistic content sharing," *2017 IEEE International Conference on Communications (ICC)*, pp. 1–6, 2017.
- [6] M. Lourenço, T. S. Gomides, F. S. H. de Souza, R. I. Meneguette, and D. L. Guidoni, "A traffic management service based on v2i communication for vehicular ad-hoc networks," in *Proceedings of the 10th Latin America Networking Conference - LANC '18*. ACM Press, 2018.
- [7] C. Brennand, A. Boukerche, R. Meneguette, and L. Villas, "A novel urban traffic management mechanism based on fog," *IEEE Symposium on Computers and Communications (ISCC)*, 07 2017.
- [8] D. L. Guidoni, G. Maia, F. S. H. Souza, L. A. Villas, and A. A. F. Loureiro, "Vehicular traffic management based on traffic engineering for vehicular ad hoc networks," *IEEE Access*, vol. 8, pp. 45 167–45 183, 2020.
- [9] J. Pan, I. S. Popa, and C. Borcea, "Divert: A distributed vehicular traffic re-routing system for congestion avoidance," *IEEE Transactions on Mobile Computing*, vol. 16, no. 1, pp. 58–72, 2017.
- [10] S. Wang, S. Djahel, and J. Mcmanis, "An adaptive and vanets-based next road re-routing system for unexpected urban traffic congestion avoidance," 12 2015.
- [11] J. Pan, M. A. Khan, I. S. Popa, K. Zeitouni, and C. Borcea, "Proactive vehicle re-routing strategies for congestion avoidance," in *2012 IEEE 8th International Conference on Distributed Computing in Sensor Systems*, May 2012, pp. 265–272.
- [12] T. S. Gomides, M. L. Fernandes, F. Sumika Hojo de Souza, L. Villas, and D. L. Guidoni, "Fire-nrd: A fully-distributed and vanets-based traffic management system for next road decision," in *2019 15th International Conference on Distributed Computing in Sensor Systems (DCOSS)*, May 2019, pp. 554–561.
- [13] HCM, *HCM 2010 : Highway Capacity Manual. Chapter 36, Travel Time Reliability*. National Research Council. Transportation Research Board., 2010. [Online]. Available: <https://trrjournalonline.trb.org/doi/abs/10.5555/9780309160773.41>
- [14] P. A. Lopez, M. Behrisch, L. Bieker-Walz, J. Erdmann, Y.-P. Flötteröd, R. Hilbrich, L. Lücken, J. Rummel, P. Wagner, and E. Wießner, "Microscopic traffic simulation using sumo," in *The 21st IEEE International Conference on Intelligent Transportation Systems*, 2018.
- [15] A. Varga and R. Hornig, "An overview of the omnet++ simulation environment," in *the 1st International Conference on Simulation Tools and Techniques for Communications, Networks and Systems & Workshops*, ser. Simutools '08, 2008, pp. 60:1–60:10.
- [16] C. Sommer, R. German, and F. Dressler, "Bidirectionally Coupled Network and Road Traffic Simulation for Improved IVC Analysis," *IEEE Transactions on Mobile Computing*, vol. 10, no. 1, pp. 3–15, 2011.
- [17] OpenStreetMap contributors, "Planet dump retrieved from <https://planet.osm.org>," <https://www.openstreetmap.org>, 2017.